

INTERNATIONAL LAW IMPLICATIONS OF THE DETECTION OF EXTRATERRESTRIAL INTELLIGENT SIGNALS

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Abstract—This paper first considers whether the present law of outer space, as it has been enshrined in five United Nations treaties and other legal documents concerning outer space, provides a satisfactory basis for SETI/CETI activities. In the author's opinion, these activities may serve "the common interest of all mankind in the progress of the exploration and use of outer space for peaceful purposes," as recognized in the 1967 Outer Space Treaty. The use of the radio frequency spectrum for SETI/CETI purposes should be in conformity with the legal principles governing this valuable natural resource, as expressed in the International Telecommunication Convention and related documents, and with allocations of the relevant segments of the spectrum by the competent bodies of the International Telecommunication Union.

In the second part the author examines the impact that the detection of extraterrestrial intelligent signals may have on the present body of space law. A possible role for the United Nations in this respect is also explored and a timely interest of the world body in discussing questions relating to this subject is recommended. Consideration of these questions could become a tool helping to concentrate the attention of the world community on problems of common concern and thus to strengthen international cooperation. However, the author believes that a law-making process that would aim at elaborating a special regulation of activities in this field would be premature at this stage. It should be initiated only when the boundary between possibilities and realities is crossed.

Finally, the paper outlines some likely transformation in our space law thinking that would be the consequence of the detection of extraterrestrial intelligent signals. Elaboration of the principles and norms to govern relations between the international community of our own planet and other intelligent communities in the universe would add a new dimension to the present body of outer space law. At the same time this new approach might exercise a beneficial influence on relations between nations and peoples of the planet Earth.

Considerations of legal implications of new phenomena in the world for our life are usually made from two points of view that reflect two basic levels of our thinking. First, we try to establish what exists and then decide what could or should be done. This two-fold approach, known in legal terminology as *de lege lata* and *de lege ferenda*, can also be used in the consideration of our present problem, though it seems to be still a rather extraordinary subject.

1. PRESENT STATE OF SPACE LAW AND ITS RELEVANCE TO CETI/SETI ACTIVITIES

It is generally known that during a quarter of a century following the first appearance of man in the space environment, the law of outer space developed into a relatively comprehensive set of principles and norms. It is now enshrined in five international treaties and some other documents worked out in the United Nations[1]. Though individual treaties have enjoyed so far a different degree of adherence from the part of Member States of the present world community, it can be said in general that the necessary tools for a broad international cooperation in the scientific and technological as well as the legal aspects of the exploration and peaceful uses of outer space have been provided. Can it be also affirmed that the existing legal instruments provide a satisfactory basis for CETI/SETI research, are the principles and norms enshrined in these instruments applicable to, and do they facilitate international cooperation in these particular kinds of activities?

In our opinion, CETI/SETI activities may serve "the common interest of all mankind in the progress of the exploration and use of outer space for peaceful purposes" as recognized in a

preambular paragraph of the 1967 Outer Space Treaty as one of the basic considerations that led the States Parties to this Treaty to its conclusion. Moreover, international cooperation in the CETI/SETI field may effectively contribute to one of the main aims of its drafters - the development of mutual understanding and the strengthening of friendly relations between States and peoples.

Having analyzed the 1967 Outer Space Treaty and other space law instruments in the light of CETI/SETI, we can say that at least some of these principles and norms provide guidelines that are applicable to these activities, though the directives given by them have been spelled out in rather general terms. Thus already the first paragraph of Article I of the Outer Space Treaty provides that "the exploration and use of outer space, including the moon and other celestial bodies, shall be carried out for the benefit and in the interests of all countries, irrespective of their degree of economic or scientific development, and shall be the province of all mankind." CETI/SETI activities shall therefore be considered as a part of space exploration that has been qualified as "the province of all mankind." It has been correctly stated in the Report of the Second United Nations

Conference on the Exploration and Peaceful Uses of Outer Space in a short paragraph dealing with the search for extraterrestrial intelligence that this quest "is of relevance to all mankind" and that "it should therefore be considered a joint international effort, with each country contributing whatever it can[2]."

Attention should also be paid to Article IX of the Outer Space Treaty which emphasizes the principle of cooperation and mutual assistance in the exploration and use of outer space, including the Moon and other celestial bodies and due regard to the corresponding interests of other States Parties to the Treaty in the conduct of such activities. Finally, at least Article XI of the same Treaty, which is particularly relevant to possible detections of ETI signals and the dissemination of such observations, should be recalled. In this provision States Parties to this Treaty agreed to inform the Secretary General of the United Nations as well as the public and the international scientific community, to the greatest extent feasible and practicable, of the nature, conduct, locations and results of their activities in outer space relating to its peaceful exploration and use. According to the same Article, the Secretary General of the United Nations, on receiving the said information, should be prepared to disseminate it immediately and effectively.

From among the other space law instruments two treaties are also relevant. First is the 1975 Convention on Registration of Objects Launched into Outer Space which provides, *inter alia*, for a United Nations Register in which information furnished by States on each space object carried on their national registries should be recorded. Such information shall also specify the general function of each space object. Thus, when space objects are used for purposes of CETI/SETI activities their function in this particular field should be adequately characterized in the information furnished to the United Nations Register and, through it, the world community will get at least general information about such on-going activities.

Second, the 1979 Agreement Governing the Activities of States on the Moon and Other Celestial Bodies should be also recalled in this connection. In this Agreement the above mentioned principles of the 1967 Outer Space Treaty have been elaborated in relation to the specific conditions of the moon and other celestial bodies within the solar system. One of its provisions, Article 5, para. 3, deals specifically with "organic life" in outer space in the following way: "In carrying out activities

under this Agreement, States Parties shall promptly inform the Secretary General, as well as the public and the international scientific community, of any phenomena they discover in outer space, including the moon, which could endanger human life or health, as well as of any indication of organic life." Therefore, this provision offers a guideline that is applicable to CETI/SETI activities when developed in connection with the exploration of the moon and other celestial bodies of our solar system.

The picture of the up-to-date legal regulation of space activities would not be complete without emphasizing that the use of the radio frequency spectrum for CETI/SETI purposes, and specific arrangements, therefore should be made in conformity with the legal principles governing this valuable natural resource as expressed in the International Telecommunication Convention and related documents[3]. Allocation of the relevant segments of the spectrum for the purposes of space exploration by the competent bodies of the International Telecommunication Union has already been done. The radio regulations in force in which the bands 1400 - 1427 MHz have been allocated to Earth exploration - satellites, radio astronomy and space research, also include a footnote (722) which states: "In the bands 1400 - 1727 MHz, 101 - 120 GHz and 197 - 220 GHz, passive research is being conducted by some countries in a programme for the search for intentional emissions of extra-terrestrial origin[4]."

2. THE QUESTION OF FURTHER STEPS TO BE TAKEN

In the past quarter of a century the United Nations has become a center for harmonizing the actions of nations in the attainment of their common ends in the field of outer space. As stated in the report of UNISPACE 82, "...the United Nations, along with its specialized and associated agencies, has been playing an important role in stimulating international cooperation and in helping to promote the utilization of space technology by all countries[5]." Therefore the United Nations appears as the most competent center for developing an appropriate legal basis for CETI/SETI activities and the need for its timely interest in questions relating to this subject cannot be underestimated. Consideration of this subject could become a tool in helping to draw the attention of the world community to problems of their common concern, and thus

strengthen international cooperation and diminish mutual differences.

The question, however, is when and under what heading should such a consideration start and in this respect, a prudent approach is advisable. While it may be admitted that the present legal basis of space activities remains rather general in many respects, it would not be wise to request an immediate consideration of a special regulation governing the CETI/SETI activities. The problem seems far from being ripe for an early initiation of this step. The consideration of a legislative process can only be started when the boundary between possibilities and well-established realities has been crossed. Similarly as in other areas of space exploration, a close cooperation between scientists and lawyers is desirable and necessary. Lawyers will depend on the advice of scientists as to when and to what degree the emerging problems of our relations with ETI should be considered as real and urgent.

However, the attention of the world community should be drawn to these problems through the United Nations now and all progress reached in this field should be reported. This aim could be attained by different means, one of them being reports of individual nations on their space activities which are submitted from time to time to the Committee on the Peaceful Uses of Outer Space (COPUOS) and published as documents of this United Nations body. Similarly, the reports presented on an annual basis on the progress in space research and the development of space technology by COSPAR and the IAF respectively[6], may serve as a source of information on CETI/SETI activities. For example this year's report of the IAF might refer in some greater detail to the results of this specialized meeting held during the XXXVIIIth International Astronautical Federation Congress.

Moreover, the problems of ETI might be brought to the attention of the United Nations through the Scientific and Technical Sub-Committee of COPUOS which will begin, at its next session, in February 1987, to consider two new points on its agenda, namely "matters relating to planetary exploration" and "matters relating to astronomy[7]."

In this connection, the role of international, non-governmental organizations, such as the IAF and COSPAR, should be appreciated. They offer useful and appropriate fora for mutual exchange of information on the nature of the problems and progress reached in their

solutions. The observer status of both these non-governmental scientific organizations within COPUOS offers a good basis for close cooperation with the United Nations in this field.

Finally, let me emphasize one thing. My advice concerning the possible initiation of a legislative process does not mean that interest in, and studies of, legal aspects of CETI/SETI would be premature. On the contrary, the examination of these aspects on an international level which - as evidenced by this meeting - has already started, should be encouraged. Such studies help us to prepare our legal thinking for its necessary adjustment in the future. For the task of elaborating special principles and norms to govern relations between the international community of our own planet and other intelligent communities in the universe, if and when they are discovered and the relations with them established, would require adding a new dimension to the law of outer space. At the same time, this new approach will exercise a beneficial influence on relations between nations and peoples of the planet Earth.

REFERENCES

1. See their texts in *Space Activities of the United Nations and International Organizations*. Annex I, United Nations, New York, pp. 199-232 (1986).
2. Report of the Second United Nations Conference on the Exploration and Peaceful Uses of Outer Space. *UNISPACE 82*, Vienna, doc. A/CONF.101/10, para. 310 at p. 76 (9-21 August 1982).
3. See International Telecommunication Convention. Final Protocol, Additional Protocols, Optional Additional Protocol, Resolutions, Recommendation and Opinions, Nairobi (1982). General Secretariat of the International Telecommunication Union, Geneva.
4. See Radio Regulation 8-94 and also Report 700-1 on Radiocommunication requirements for systems to search for extraterrestrial intelligence (SETI).
5. See the source listed under ref. 2, para. 381 at p. 92.
6. *Progress of Space Research 1984-85*, report submitted by the Committee on Space Research (COSPAR) of the International Council of Scientific Unions (ICSU), doc. A/AC.105/C.1/146 (3 January 1986) and *Highlights in Space Technology 1985*, report

submitted by the International Astronautical Federation (IAF), doc. A/AC.105/C.1/147 (3 February 1986).

7. Report of the Scientific and Technical Sub-Committee on the Work of Its Twenty-Third Session, para. 99 (e) and (f) at 21, doc. A/AC.105/369 (28 February 1986).

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